THE FRESHWATER MUSSEL RESOURCES IN A SELECTED SEGMENT OF THE SALINE RIVER:

LOCATION, SPECIES COMPOSITION AND STATUS OF MUSSEL BEDS

ADDENDUM 2



Saline River Mussel Bed SRB146 approximately 6 river miles downstream from Johnsville, Arkansas

Prepared for the
Arkansas Game and Fish Commission

November, 2004

THE FRESHWATER MUSSEL (BIVALVIA:UNIONACEA) RESOURCES IN A SELECTED SEGMENT OF THE SALINE RIVER: LOCATION, SPECIES COMPOSITION AND STATUS OF MUSSEL BEDS

ADDENDUM 2
ARKANSAS HIGHWAY 15 TO FELSENTHAL NATIONAL WILDLIFE REFUGE

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Prepared for the
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Russellville, Arkansas
November, 2004

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BACKGROUND

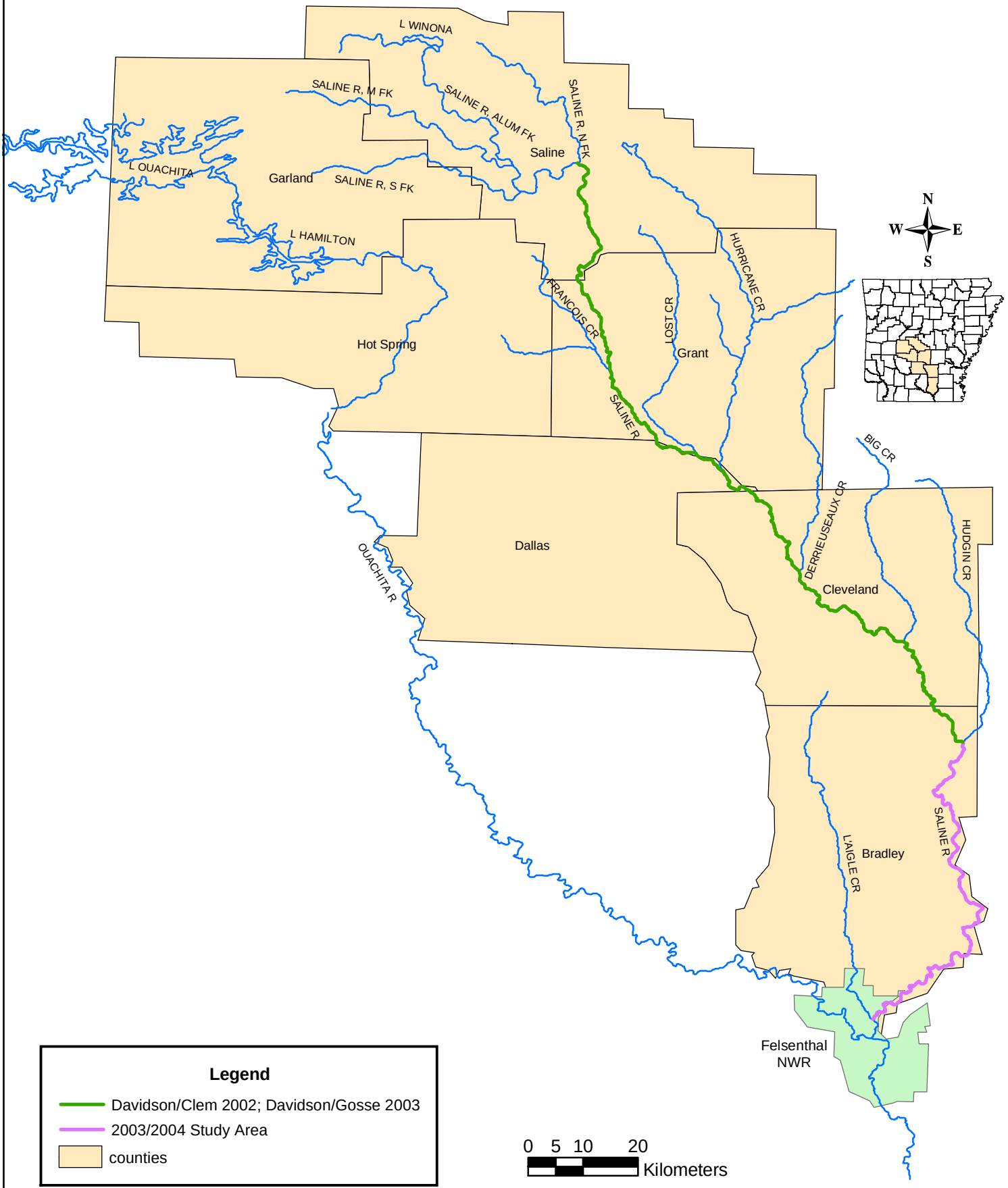
This addendum completes a multi-year survey of the main stem Saline River from near Tull, Arkansas to the northern boundary of Felsenthal National Wildlife Refuge (NWR). The survey began in 2001 with funding provided by The Nature Conservancy's Arkansas Field Office and the Arkansas Game and Fish Commission (AGFC). A final report was prepared in December, 2002 for surveys of the Saline River from Tull, Arkansas to Arkansas Highway 15 near Warren, Arkansas (Davidson and Clem 2002). At that time, no funding was available to survey the Saline River from Arkansas Highway 15 to Felsenthal NWR. Subsequently, the AGFC provided funding to complete this reach of the Saline River in 2003 and 2004. Results from this survey are being submitted as an addendum to the December, 2002 report.

Gordon et al. (1979) reviewed literature and museum specimens on Arkansas' unionacean fauna and subsequently listed 35 species as known or possible elements of the Saline River unionacean fauna. Headwater reaches of the Saline River to the Saline/Grant County line, Arkansas have been adequately surveyed (Harris and Gordon 1988; Brown and Brown 1989; U.S. Fish and Wildlife Service 1990; U.S. Fish and Wildlife Service 1992; Burns & McDonnell 1992a, 1992b; Davidson and Gosse 2003). Davidson (1997) surveyed the lower 18 km of the Saline River that lies within the boundary of Felsenthal NWR. Davidson and Clem (2002) reported 42 species in the Saline River, an additional 4 species that may occur in backwater areas, and genetic analyses that may add an additional 4 species. A few unionid records exist for highway crossings (John Harris, AHTD) on the Saline River. Otherwise, species composition, abundance and mussel bed locations are unknown for an 81 km (50.6 rm) section of the Saline River from Arkansas Highway 15 to the northern boundary of Felsenthal NWR.

The major objective of this study was to identify federally listed threatened and endangered species of freshwater mussels, as well as global and state species of concern. Additional objectives include; documentation of species composition, approximate abundance and location of mussel beds. Results will provide baseline information on the status of the mussel fauna inhabiting the lower Saline River. The data derived from this survey will include: mussel bed dimensions, water depth, habitat type, mapped locations of mussel beds, any apparent stressors; and biological data such as species composition, and approximate abundance.

Methods and procedures used during this survey are outlined in Davidson and Clem (2002). The Saline River from Arkansas Highway 15 to the northern boundary of Felsenthal NWR was accessed using a motorized flat bottom boat. Approximately 81 km (50.6 rm) were surveyed during low water conditions between August to September 2003 and August to September 2004 (Figure A1).

Figure A1. Saline River Unionid Study Areas, 2001-2004.



Davidson and Clem, 2004

RESULTS

Mussel beds and concentrations were generally located in three physical settings: 1) riffle/run areas; 2) along descending banks immediately upstream of riffle/run areas; or 3) in river bends. Mussels were located in water depths, at low flow conditions, ranging from approximately 0.2 m to 5 m. Mussels were most often associated with gravel/sand substrate. Pools corresponding with long straight stretches and reduced water velocity generally were devoid of mussels or major aggregations of mussels.

A total of 74 mussel beds and 9 mussel concentrations were defined in the 81 km (50.6 rm) surveyed (Table A1; Figure A2). Addendum one, a series of U. S. Geological Survey topographic maps with mussel beds and concentrations marked and labeled, was provided to the AGFC (Davidson and Clem 2002). Thirty-five species were found within the study area (Table A2). Of these, two species are listed by the U. S. Fish and Wildlife Service (USFWS) as federally endangered, five species are listed by The Nature Conservancy (TNC) as G1 or G2 and eight species are listed by the Arkansas Natural Heritage Commission (ANHC) as S1 or S2. Conservation status by the USFWS, TNC and ANHC is summarized in Davidson and Clem (2002). *Quadrula apiculata* was the only species found in this section of the Saline River that has not been reported from previous surveys.

Estimated average mussel densities in defined mussel beds down stream of Arkansas Highway 15 ranged from 12 to 105 mussels/m^2. Estimated mussel densities in beds downstream of Arkansas Highway 15 often exceeded 100 mussels/m^2, although estimated mean bed densities seldom exceeded 60 mussels/m^2. Mussel concentration area ranged from 40 to 600 m^2. Seventeen mussel beds were less than 500 m^2, 11 were between 500 to 1,000 m^2, 37 between 1,000 to 5,000 m^2, 6 between 5,000 to 10,000 m^2, and 3 greater than 10,000 m^2. The largest beds were located downstream of SRB141. The largest bed (SRB159) was 18,170 m^2. The next three largest beds were SRB149 (17,400 m^2), SRB147 (11,830 m^2) and SRB141 (9,600 m^2; Table A3).

Community Composition

A total of 10,112 individuals were examined. *Elliptio dilatata* (31.3 percent) was the dominant species within the survey area. Other common species included *Quadrula pustulosa* (14.5 percent), *Actinonaias ligamentina* (14.0 percent), *Fusconaia/Pleurobema* type (7.5 percent), and *Amblema plicata* (5.3 percent). Federally threatened and endangered species, TNC G1 and G2 species and ANHC S1 and S2 species accounted for 0.2, 8.7 and 9.0 percent of the total community, respectively. *Elliptio dilatata* was (co)dominant in 61 of 83 mussel beds and concentrations between Arkansas Highway 15 and Felsenthal NWR. Other dominant species included *Actinonaias ligamentina* (10), *Quadrula pustulosa* (10), *Amblema plicata* (2) and *Obovaria jacksoniana* (1; Table A2).

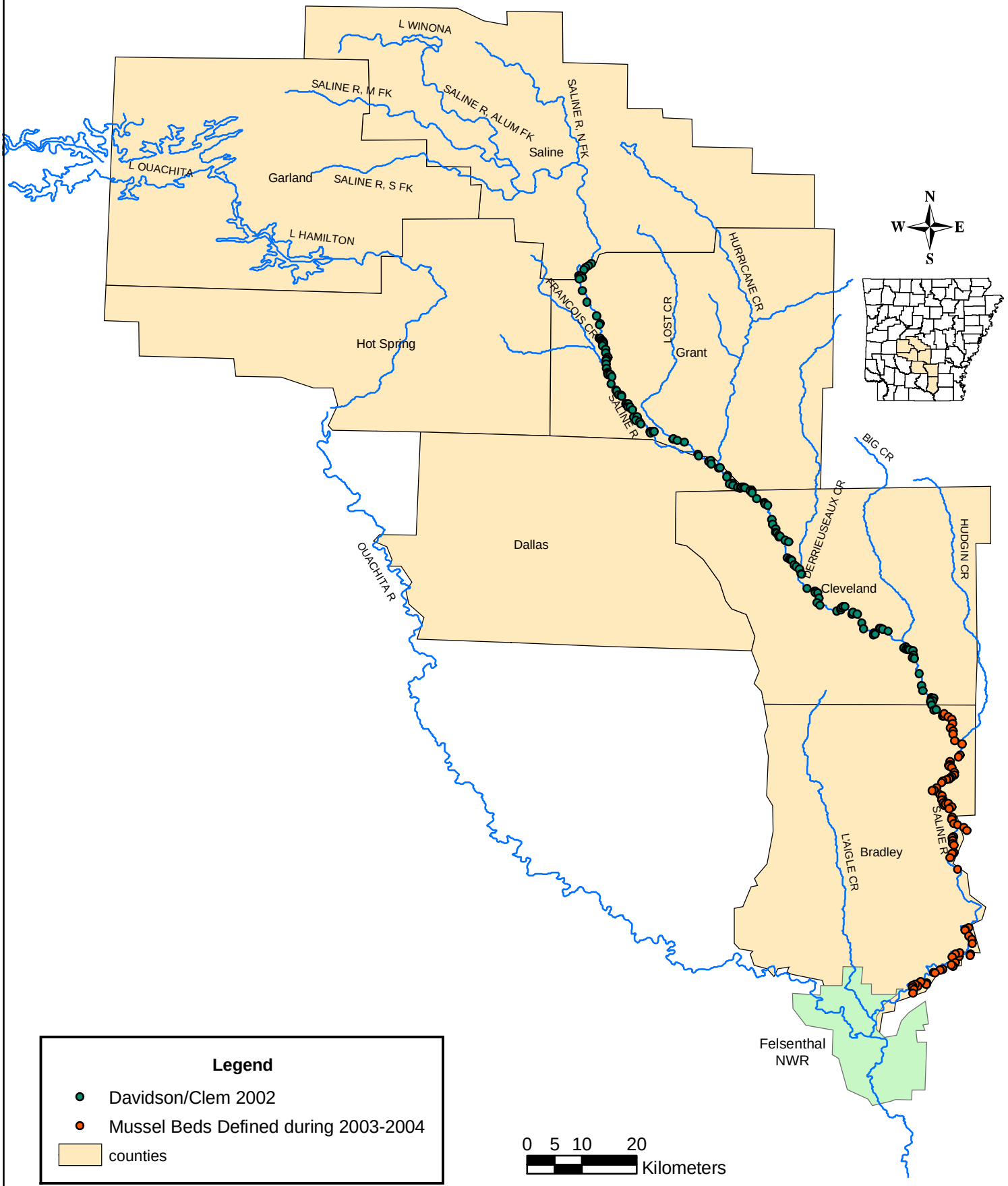
Table 1. Locational data for mussel beds and concentrations in the Saline River from Arkansas Highway 15 near Warren to the northern boundary of Felsenthal National Wildlife Refuge.

Site ID	Latitude	Longitude	Quad	Section	Township	Range
SRC53	33.694644	-92.023822	Warren NE	NW1/4 SW1/4 3	12S	9W
SRC54	33.694128	-92.021681	Warren NE	NE1/4 SW1/4 3	12S	9W
SRC55	33.697925	-92.020858	Warren NE	SE1/4 NW1/4 3	12S	9W
SRB96	33.693019	-92.014569	Warren NE	SE1/4 SE1/4 3	12S	9W
SRB97	33.688522	-92.009053	Warren NE	NW1/4 NW1/4 11	12S	9W
SRB98	33.681758	-92.007994	Warren NE	NW1/4 SW1/4 11	12S	9W
SRB99	33.674753	-92.011011	Warren NE	NE1/4 NE1/4 15	12S	9W
SRB100	33.669442	-92.007058	Warren NE	SW1/4 NW1/4 14	12S	9W
SRB101	33.664739	-92.007172	Warren NE	SW1/4 SW1/4 14	12S	9W
SRB102	33.652956	-92.003958	Warren NE	SE1/4 NW1/4 23	12S	9W
SRC56	33.647853	-91.991811	Wilmar North	SW1/4 SW1/4 24	12S	9W
SRB103	33.630011	-91.994711	Wilmar North	NE1/4 NE1/4 35	12S	9W
SRB104	33.627033	-91.997556	Wilmar North	SW1/4 NE1/4 35	12S	9W
SRB105	33.618842	-92.012003	Warren	SE1/4 SE1/4 34	12S	9W
SRB106	33.614280	-92.012120	Warren	NW1/4 NW1/4 2	13S	9W
SRB107	33.612840	-92.013850	Warren	SE1/4 NE1/4 3	13S	9W
SRB108	33.611140	-92.013100	Warren	SE1/4 NE1/4 3	13S	9W
SRB109	33.609470	-92.009350	Warren	SW1/4 NW1/4 2	13S	9W
SRB110	33.601100	-92.005010	Warren	NW1/4 NW1/4 11	13S	9W
SRB111	33.597340	-92.005060	Warren	SE1/4 NW1/4 11	13S	9W
SRC57	33.594380	-92.009350	Warren	NW1/4 NW1/4 11	13S	9W
SRB112	33.591990	-92.012040	Warren	NE1/4 SE1/4 10	13S	9W
SRC58	33.590690	-92.013190	Warren	SE1/4 SE1/4 10	13S	9W
SRB113	33.589490	-92.016620	Warren	SW1/4 SE1/4 10	13S	9W
SRB114	33.587110	-92.023360	Warren	SE1/4 SW1/4 10	13S	9W
SRB115	33.584460	-92.025290	Warren	NW1/4 NW1/4 15	13S	9W
SRB116	33.575730	-92.034010	Warren	SW1/4 SE1/4 16	13S	9W
SRB117	33.571660	-92.037490	Warren	NW1/4 NE1/4 21	13S	9W
SRB118	33.571190	-92.040760	Warren	NE1/4 NW1/4 21	13S	9W
SRB119	33.564480	-92.028070	Warren	NE1/4 SE1/4 21	13S	9W
SRC59	33.562880	-92.025050	Warren	NW1/4 SW1/4 22	13S	9W
SRB120	33.555628	-92.022647	Warren	SE1/4 NW1/4 27	13S	9W
SRB121	33.555600	-92.025397	Warren	SW1/4 NW1/4 27	13S	9W
SRB122	33.551281	-92.023414	Warren	NE1/4 SW1/4 27	13S	9W
SRB123	33.547158	-92.021189	Warren	SE1/4 SW1/4 27	13S	9W
SRC60	33.549342	-92.018892	Warren	NW1/4 SE1/4 27	13S	9W
SRC61	33.550158	-92.014461	Warren	NE1/4 SE1/4 27	13S	9W
SRB124	33.544311	-92.009150	Warren	SW1/4 SW1/4 26	13S	9W
SRB125	33.541992	-92.013239	Warren	NE1/4 NE1/4 34	13S	9W
SRB126	33.527842	-92.008322	Warren	NW1/4 NW1/4 2	14S	9W
SRB127	33.523942	-92.008783	Warren	NW1/2 2	14S	9W
SRB128	33.516558	-92.005403	Warren	SE1/4 SW1/4 2	14S	9W
SRB129	33.514556	-91.999036	Wilmar South	S2/S11	14S	9W
SRB130	33.510661	-91.989522	Warren	SE1/4 NW1/4 11	14S	9W
SRB131	33.505817	-91.984442	Warren	NW1/4 SE1/4 11	14S	9W
SRB132	33.495367	-92.006564	Sumpter	SW1/4 NW1/4 14	14S	9W
SRB133	33.493194	-92.008175	Sumpter	NW1/4 SW1/4 14	14S	9W
SRB134	33.489394	-92.007597	Sumpter	SW1/4 SW1/4 14	14S	9W
SRB135	33.484458	-92.007594	Sumpter	NW1/4 NW1/4 23	14S	9W
SRB136	33.481547	-92.005703	Sumpter	SW1/4 NW1/4 23	14S	9W
SRB137	33.469660	-92.005810	Sumpter	NW1/4 NW1/4 26	14S	9W
SRB138	33.467710	-92.008730	Sumpter	NW1/4 NW1/4 26	14S	9W

Table 1. Locational data for mussel beds and concentrations in the Saline River from Arkansas Highway 15 near Warren to the northern boundary of Felsenthal National Wildlife Refuge.

Site ID	Latitude	Longitude	Quad	Section	Township	Range
SRB139	33.460450	-92.012320	Sumpter	NE1/4 SE1/4 27	14S	9W
SRB140	33.441290	-91.999130	Longview	NE1/4 NW1/4 2	15S	9W
SRB141	33.345970	-91.980570	Milo	NW1/4 SE1/4 1	16S	9W
SRB142	33.342860	-91.986950	Milo	SE1/4 SW1/4 1	16S	9W
SRB143	33.341320	-91.987000	Milo	SE1/4 SW1/4 1	16S	9W
SRB144	33.332490	-91.980420	Milo	NE1/4 SE1/4 12	16S	9W
SRB145	33.326780	-91.975920	Milo	SW1/4 SW1/4 7	16S	8W
SRB146	33.326790	-91.975910	Milo	NW1/4 NW1/4 8	16S	8W
SRB147	33.319180	-91.974730	Milo	W1/2 18	16S	8W
SRB148	33.302560	-91.977910	Milo	NE1/4 SE1/4 24	16S	9W
SRB149	33.300680	-91.978910	Milo	SE1/4 24	16S	9W
SRB150	33.298630	-91.996140	Milo	SE1/4 SE1/4 23	16S	9W
SRB151	33.304620	-91.995470	Milo	NE1/4 SE1/4 23	16S	9W
SRB152	33.302790	-92.002560	Vick	NW1/4 SE1/4 23	16S	9W
SRB153	33.297490	-92.008600	Vick	SW1/4 SW1/4 23	16S	9W
SRB154	33.288190	-92.002440	Vick	NE1/4 SW1/4 26	16S	9W
SRB155	33.282950	-92.007250	Vick	SW1/4 26/NW1/4 35	16S	9W
SRB156	33.284980	-92.010130	Vick	SW1/4 SW1/4 26	16S	9W
SRB157	33.277870	-92.023250	Vick	SE1/4 NW1/4 34	16S	9W
SRB158	33.276230	-92.028160	Vick	SW1/4 NW1/4 34	16S	9W
SRB159	33.271660	-92.039120	Vick	W1/2 33	16S	9W
SRB160	33.271050	-92.036510	Vick	SW1/4 SE1/4 33	16S	9W
SRB161	33.254767	-92.050414	Vick	SE1/4 SE1/4 5	17S	9W
SRB162	33.252997	-92.050842	Vick	NW1/4 NE1/4 8	17S	9W
SRB163	33.257175	-92.059911	Vick	SW1/4 SW1/4 5	17S	9W
SRB164	33.249703	-92.064292	Vick	SE1/4 NE1/4 7	17S	9W
SRB165	33.252542	-92.069078	Vick	NW1/4 NE1/4 7	17S	9W
SRB166	33.250336	-92.073769	Vick	SE1/4 NE1/4 7	17S	9W
SRB167	33.247064	-92.073508	Marais Saline	NE1/4 SW1/4 7	17S	9W
SRB168	33.244428	-92.070683	Marais Saline	NW1/4 SE1/4 7	17S	9W
SRB169	33.238153	-92.072750	Marais Saline	NE1/4 NW1/4 18	17S	9W

Figure A2. Freshwater Mussel (Unionacea) bed and concentration locations in the Saline River from near Tull, Arkansas to Felsenthal NWR, 2001-2004.



Davidson and Clem, 2004

Table 2. Mussel bed and concentration data for the Saline River from Arkansas Highway 15 to Felsenthal NWR, Arkansas.

Scientific Name	SRC53	SRC54	SRC55	SRB96	SRB97	SRB98	SRB99	SRB100	SRB101	SRB102	SRC56	SRB103	SRB104	SRB105	SRB106
Actinonaias ligamentina	11	21	12	23	12	10	64	19	14	14	15	37	4	16	29
Alasmidonta marginata									1						
Amblema plicata						4	1	3	7	1	1	5	1	3	1
Cyprogenia aberti		3	5	1	4	2	8	2	13	4	1	11	2	5	11
Ellipsaria lineolata		1	8	1	1							1	1		
Elliptio dilatata	23	10		33	9	18	99	47	21	12	10	62	16	21	27
Fusconaia ebena		1			1		1	1			1	2		4	1
Fusconaia flava		1			2	4	1	5	14	2		15	5		4
Fusconaia/Pleurobema type	2	5	1		1	10	26	10	16	4	2	19	5	5	14
Lampsilis abrupta															
Lampsilis cardium		1	1	3	1		3		1			1		1	2
Lampsilis hydiana	1														
Lampsils powelli															
Lampsilis teres								1							
Lasmigona costata															
Leptodea fragilis				1	1										
Ligumia recta									1		1	2			1
Megalonaias nervosa												1			
Obliquaria reflexa		2							2						1
Obovaria jacksoniana	1	2								1				1	1
Plectomerus dombeyanus									1						
Pleurobema rubrum				1	2	4	14	2	6	1		20	2	3	6
Potamilus purpuratus		1		1											
Ptychobranchus occidentalis		2	1	2	3		5	R	6	1	1	3	1	3	2
Quadrula cylindrica							2	1							
Quadrula apiculata									1						
Quadrula fragosa									1						
Quadrula metanevra	1			2	1		1	2		3		8		6	2
Quadrula nodulata															
Quadrula pustulosa	10	15	2	3	4	18	6	12	23	13	3	15	8	13	13
Quadrula quadrula															
Strophitus undulatus	1				1							1			
Toxolasma lividus									R						
Tritogonia verrucosa							1	1	2	1		6	1	1	
Truncilla donaciformis															
Truncilla truncata					1		4	1				2			
Villosa lienosa															
Number of Live Individuals	50	65	30	71	44	70	236	107	130	57	35	211	46	82	115
Number of relicts	NC	NC	NC	NC	NC	NC	NC	1	1	NC	NC	NC	NC	NC	NC
Number of Species	8	12	7	11	15	8	15	15	18	12	9	18	11	13	15
Area (m^2)	600	NC	300	300	1200	200	4000	460	1470	360	40	1400	440	500	2160
Approximate Density (#/m^2)	8	12	8	25	20	18	50	14	40	35	35	35	13	23	35

Table 2. Mussel bed and concentration data for the Saline River from Arkansas Highway 15 to Felsenthal NWR, Arkansas.

Scientific Name	SRB107	SRB108	SRB109	SRB110	SRB111	SRC57	SRB112	SRC58	SRB113	SRB114	SRB115	SRB116	SRB117	SRB118
Actinonaias ligamentina	40	16	12	32	34	100	17	11	30	61	13	30	36	39
Alasmidonta marginata				1						1				
Amblema plicata	1	2	2	3	3	10			3	18	2	14	2	6
Cyprogenia aberti	8	5	5	20	4			4	3	19	1	15	4	9
Ellipsaria lineolata		1		1			1		1	4		2		4
Elliptio dilatata	54	26	15	43	13	30	43	20	34	65	14	24	47	70
Fusconaia ebena		1	2		1	1				5		1		
Fusconaia flava	1	7	7	9		3	1	2	16	25	5	9	5	6
Fusconaia/Pleurobema type	8	16	5	10	2	10	6	7	26	37	7	8	13	16
Lampsilis abrupta														
Lampsilis cardium	1			1		1		1		2		1	5	9
Lampsilis hydiana				1										R
Lampsils powelli														
Lampsilis teres												R		2
Lasmigona costata														
Leptodea fragilis		1											1	1
Ligumia recta						1		R		R		R		
Megalonaias nervosa				1										
Obliquaria reflexa			1		2					5		8		1
Obovaria jacksoniana		1		1	1		1		1	3		1		4
Plectomerus dombeyanus				1					1	2		2		1
Pleurobema rubrum	6		2	4	2	10		2	15	23	1	12	2	5
Potamilus purpuratus								R	1			R		7
Ptychobranchus occidentalis		4	1	12	4		1		3	14	2	5	3	3
Quadrula cylindrica													1	2
Quadrula apiculata														
Quadrula fragosa														
Quadrula metanevra	6	1	3	17	10	6	3	1	3	12	1	2	8	13
Quadrula nodulata														
Quadrula pustulosa	7	11	10	18	13	20	4	6	14	26	6	9	9	25
Quadrula quadrula														
Strophitus undulatus				1					1	1	2		1	R
Toxolasma lividus				R						R				R
Tritogonia verrucosa		2		1	R		1	1		1		3		3
Truncilla donaciformis														
Truncilla truncata			1	4						5		3		1
Villosa lienosa														
Number of Live Individuals	132	94	66	181	89	192	78	55	152	329	54	149	137	227
Number of relicts	NC	NC	NC	Numerous	1	Numerous	NC	2	NC	2	NC	Numerous	NC	14
Number of Species	10	14	13	21	13	11	10	12	15	22	11	21	14	24
Area (m^2)	160	180	140	2160	600	200	160	WS	4800	1650	400	1060	900	2850
Approximate Density (#/m^2)	25	20	20	35	30	8	23	13	35	105	15	20	20	25

Table 2. Mussel bed and concentration data for the Saline River from Arkansas Highway 15 to Felsenthal NWR, Arkansas.

Scientific Name	SRB119	SRC59	SRB120	SRB121	SRB122	SRB123	SRC60	SRC61	SRB124	SRB125	SRB126	SRB127	SRB128	SRB129	SRB130
Actinonaias ligamentina	13	16	25	16	28	14	7	19	15	6	14	13	29	20	42
Alasmidonta marginata														1	
Amblema plicata		3	3	6	8	2		1	1		5	8	10	6	3
Cyprogenia aberti	4	1	10	4	4	2	1	1	3	1	5	1	6		2
Ellipsaria lineolata				1		2	1			1	1	1	4		
Elliptio dilatata	47	21	68	32	61	31	14	30	18	49	26	37	42	61	82
Fusconaia ebena			1		1				1	1	1	1	3		1
Fusconaia flava	4		8	6	1				3	5	4	1	3	5	1
Fusconaia/Pleurobema type	7	3	37	17	37	5	5	3	8	19	15	9	14	14	15
Lampsilis abrupta													1		1
Lampsilis cardium					1	1			1		1	1	2	4	1
Lampsilis hydiana	R										1				
Lampsils powelli															
Lampsilis teres			1											1	
Lasmigona costata		1													
Leptodea fragilis	1				1						1		1	1	
Ligumia recta															
Megalonaias nervosa											1				
Obliquaria reflexa			4	1					1	2	1	1			
Obovaria jacksoniana	1					1			2	1	1	1	4	3	
Plectomerus dombeyanus				1								1			
Pleurobema rubrum	1		12	16	19		1	3	5	15	6	9	5	4	1
Potamilus purpuratus										1				1	
Ptychobranchus occidentalis			9	4	3	3		2	2	4	4	5	4	4	7
Quadrula cylindrica													1		
Quadrula apiculata															
Quadrula fragosa															
Quadrula metanevra	6	4	5	7	12	4	3		3	5	3	1	4	4	1
Quadrula nodulata															
Quadrula pustulosa	5		27	13	18	10	5	8	12	17	25	7	33	21	14
Quadrula quadrula															
Strophitus undulatus			1								1				1
Toxolasma lividus															
Tritogonia verrucosa			1						1		1			2	
Truncilla donaciformis															
Truncilla truncata				1	4			1	3		1	2	2	7	3
Villosa lienosa			1		1										
Number of Live Individuals	89	49	213	125	199	75	37	68	79	127	118	99	168	159	175
Number of relicts	1	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC
Number of Species	11	7	16	14	15	11	8	9	16	14	21	18	18	17	15
Area (m^2)	320	40	2820	3450	3120	2000	600	350	1510	2000	3750	7200	5700	3360	1520
Approximate Density (#/m^2)	23	18	50	40	50	33	9	9	23	18	28	50	40	50	45

Table 2. Mussel bed and concentration data for the Saline River from Arkansas Highway 15 to Felsenthal NWR, Arkansas.

Scientific Name	SRB131	SRB132	SRB133	SRB134	SRB135	SRB136	SRB137	SRB138	SRB139	SRB140	SRB141	SRB142	SRB143	SRB144
Actinonaias ligamentina	13	24	25	18	25	21	29	34	19	23	9	2	6	1
Alasmidonta marginata														
Amblema plicata	6	3	2	3	3		2		1		41	15	11	6
Cyprogenia aberti	2	4	7	4		3	1	3	3	4	5		2	
Ellipsaria lineolata		1	4	2	1	1	1	3		1	1	1	1	1
Elliptio dilatata	85	58	56	30	61	42	23	29	32	43	1	4	24	
Fusconaia ebena							3			2	1			1
Fusconaia flava	1	5	18	3	3	1	1	16	4	2	15	12	8	9
Fusconaia/Pleurobema type	14	26	19	5	10	3	7	10	6	5				
Lampsilis abrupta											3		1	
Lampsilis cardium	1	1	2	4				2	1	1	3		3	2
Lampsilis hydiana											1			
Lampsils powelli														
Lampsilis teres											3	1	1	
Lasmigona costata														
Leptodea fragilis				1							R			
Ligumia recta				1										
Megalonaias nervosa						1					2			
Obliquaria reflexa						1	1		1	1	22	1	1	3
Obovaria jacksoniana		1	2	1	1		1	1	3	2	18	5	6	13
Plectomerus dombeyanus											21	6	2	1
Pleurobema rubrum	2	5	12	3	3	3	6	4	2	5	1			
Potamilus purpuratus											R			1
Ptychobranchus occidentalis		4	4	6	3	2	5	2	4	1				
Quadrula cylindrica											R			
Quadrula apiculata														
Quadrula fragosa											3			
Quadrula metanevra	3	4	5	3	1	4	7	3	2	1	3	8	8	3
Quadrula nodulata														1
Quadrula pustulosa	10	10	24	17	20	5	35	11	16	6	37	52	22	13
Quadrula quadrula				1							1			1
Strophitus undulatus			1				1							
Toxolasma lividus														
Tritogonia verrucosa				1				1			28	2	4	
Truncilla donaciformis												1		
Truncilla truncata	1	2	4			1	1	1	2		2		3	3
Villosa lienosa					1									
Number of Live Individuals	138	148	185	103	132	88	124	120	96	97	221	110	103	59
Number of relicts	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	6	1	NC	NC
Number of Species	11	14	15	16	12	13	16	15	14	13	25	13	16	15
Area (m^2)	1225	2475	1890	1000	1280	600	240	2400	1350	920	9600	960	840	100
Approximate Density (#/m^2)	50	35	40	18	23	23	23	75	20	35	35	28	50	12

Table 2. Mussel bed and concentration data for the Saline River from Arkansas Highway 15 to Felsenthal NWR, Arkansas.

Scientific Name	SRB145	SRB146	SRB147	SRB148	SRB149	SRB150	SRB151	SRB152	SRB153	SRB154	SRB155	SRB156	SRB157	SRB158
Actinonaias ligamentina	3	9	26	2	20		3	2	5	8	9	4	10	3
Alasmidonta marginata														
Amblema plicata	14	7	15	13	39	1	5	6	2	3	10	10	5	6
Cyprogenia aberti	2	4	4	1	2				1			1	2	1
Ellipsaria lineolata	1	1	4	1	2		1	1		1	2			
Elliptio dilatata	13	60	76	11	128	15	25	34	102	82	90	14	74	44
Fusconaia ebena					1						2		1	
Fusconaia flava	12	11	15	4	15	1	13	13	6	7	29	5	11	15
Fusconaia/Pleurobema type	1	4	6	3	18	2	4	6	7	4	16	3	6	5
Lampsilis abrupta			1		1									
Lampsilis cardium	1	1	10	2	6	1	1	1	2		2	R	4	2
Lampsilis hydiana			4	3				1			1			
Lampsils powelli														
Lampsilis teres	R		3		R						R			
Lasmigona costata														
Leptodea fragilis			1					1			1		1	
Ligumia recta														
Megalonaias nervosa		2	3	1			1	1			10		2	
Obliquaria reflexa			6		2		2				1			
Obovaria jacksoniana	9		21	26	10	1	1	1	1	7	9	13	7	7
Plectomerus dombeyanus	1	2	7	16	11	1	1	1		1	1		3	1
Pleurobema rubrum	1		1	2	2	2		3		1	12	2	4	5
Potamilus purpuratus	2	1	3	1		1								
Ptychobranchus occidentalis			3							2	1	1	1	3
Quadrula cylindrica		9	2	R	R								1	
Quadrula apiculata					R									
Quadrula fragosa	1	2	R					1						
Quadrula metanevra	1	19	9	1	11	2		1	2		1	1	4	5
Quadrula nodulata														
Quadrula pustulosa	22	22	62	29	54	9	21	29	13	20	53	22	44	30
Quadrula quadrula	1	1	2		1		1	1	1		2	1		
Strophitus undulatus								1			1			
Toxolasma lividus														
Tritogonia verrucosa	2	3	3	3	6		1	6			1		3	2
Truncilla donaciformis					1				1				1	
Truncilla truncata	7	8	7	1	6	2	1		2	3	3	4	2	4
Villosa lienosa														
Number of Live Individuals	94	166	294	120	336	38	81	110	145	139	257	81	186	133
Number of relicts	1	1	5	1	3	NC	NC	NC	NC	NC	1	1	NC	NC
Number of Species	19	18	26	19	22	12	15	19	13	13	23	14	20	15
Area (m^2)	850	1020	11230	560	17400	300	3000	2760	1900	225	8800	600	2800	6300
Approximate Density (#/m^2)	18	55	45	33	60	23	35	30	30	23	65	23	55	50

Table 2. Mussel bed and concentration data for the Saline River from Arkansas Highway 15 to Felsenthal NWR, Arkansas.

Scientific Name	SRB159	SRB160	SRB161	SRB162	SRB163	SRB164	SRB165	SRB166	SRB167	SRB168	SRB169
Actinonaias ligamentina	2		2	2	3		1			2	2
Alasmidonta marginata											
Amblema plicata	16		21	13	18	14	11	10	15	9	32
Cyprogenia aberti											
Ellipsaria lineolata				4	2		3	1			
Elliptio dilatata	153		51	68	41	18	17	4	16	2	13
Fusconaia ebena	1			1	2	3	1				1
Fusconaia flava	27		4	8	5	4		1		4	2
Fusconaia/Pleurobema type	13		4	4	5	1	4	1	12	6	16
Lampsilis abrupta											
Lampsilis cardium	2		2	2	1					1	2
Lampsilis hydiana							1				
Lampsils powelli											
Lampsilis teres											
Lasmigona costata											
Leptodea fragilis					1						
Ligumia recta						1					1
Megalonaias nervosa	4			2		1	1		1	2	
Obliquaria reflexa			2	5	3		3			1	
Obovaria jacksoniana	12		6	2	1	4	1	1		3	2
Plectomerus dombeyanus	14		8	5	3	2	9		6	1	8
Pleurobema rubrum	6		4	2	6		15	3	2	1	1
Potamilus purpuratus	R							1			1
Ptychobranchus occidentalis	2						1				
Quadrula cylindrica				1							
Quadrula apiculata											
Quadrula fragosa											
Quadrula metanevra	6		2	1					2		6
Quadrula nodulata											
Quadrula pustulosa	51		18	19	24	16	14	7	10	13	39
Quadrula quadrula					3	1	1		1		
Strophitus undulatus							1				1
Toxolasma lividus											
Tritogonia verrucosa					2	5	3			4	
Truncilla donaciformis											
Truncilla truncata	8		1	2	2		4		1		7
Villosa lienosa											
Number of Live Individuals	317	NR	125	141	122	70	91	29	66	49	134
Number of relicts	1	NR	NC	NC	NC	NC	NC	NC	NC	NC	NC
Number of Species	16	NR	13	17	17	12	18	9	10	13	16
Area (m^2)	18170	NR	2000	2450	2800	950	1890	450	1000	300	5250
Approximate Density (#/m^2)	60	NR	23	40	35	30	30	20	20	20	30

Table A2. Species Composition in the Saline River from AR Highway 15 near Warren, AR, to Felsenthal NWR, 2003-2004.

Scientific Name	Tot Ind.	% Tot
Actinonaias ligamentina	1,416	14.00
Alasmidonta marginata	4	0.04
Amblema plicata	537	5.31
Cyprogenia aberti	270	2.67
Ellipsaria lineolata	80	0.79
Elliptio dilatata	3,164	31.29
Fusconaia ebena	53	0.52
Fusconaia flava	500	4.94
Fusconaia/Pleurobema type	755	7.47
Lampsilis abrupta	8	0.08
Lampsilis cardium	110	1.09
Lampsilis hydiana	14	0.14
Lampsilis teres	13	0.13
Lasmigona costata	1	0.01
Leptodea fragilis	16	0.16
Ligumia recta	9	0.09
Megalonaias nervosa	37	0.37
Obliquaria reflexa	88	0.87
Obovaria jacksoniana	232	2.29
Plectomerus dombeyanus	142	1.40
Pleurobema rubrum	363	3.59
Potamilus purpuratus	23	0.23
Ptychobranchus occidentalis	178	1.76
Quadrula apiculata	1	0.01
Quadrula cylindrica	20	0.20
Quadrula fragosa	8	0.08
Quadrula metanevra	303	3.00
Quadrula nodulata	1	0.01
Quadrula pustulosa	1,470	14.54
Quadrula quadrula	20	0.20
Strophitus undulates	18	0.18
Tritogonia verrucosa	110	1.09
Truncilla donaciformis	4	0.04
Truncilla truncate	141	1.39
Villosa lienosa	3	0.03
Number of Live Individuals	10,112	100.00

Federally Threatened and Endangered Species

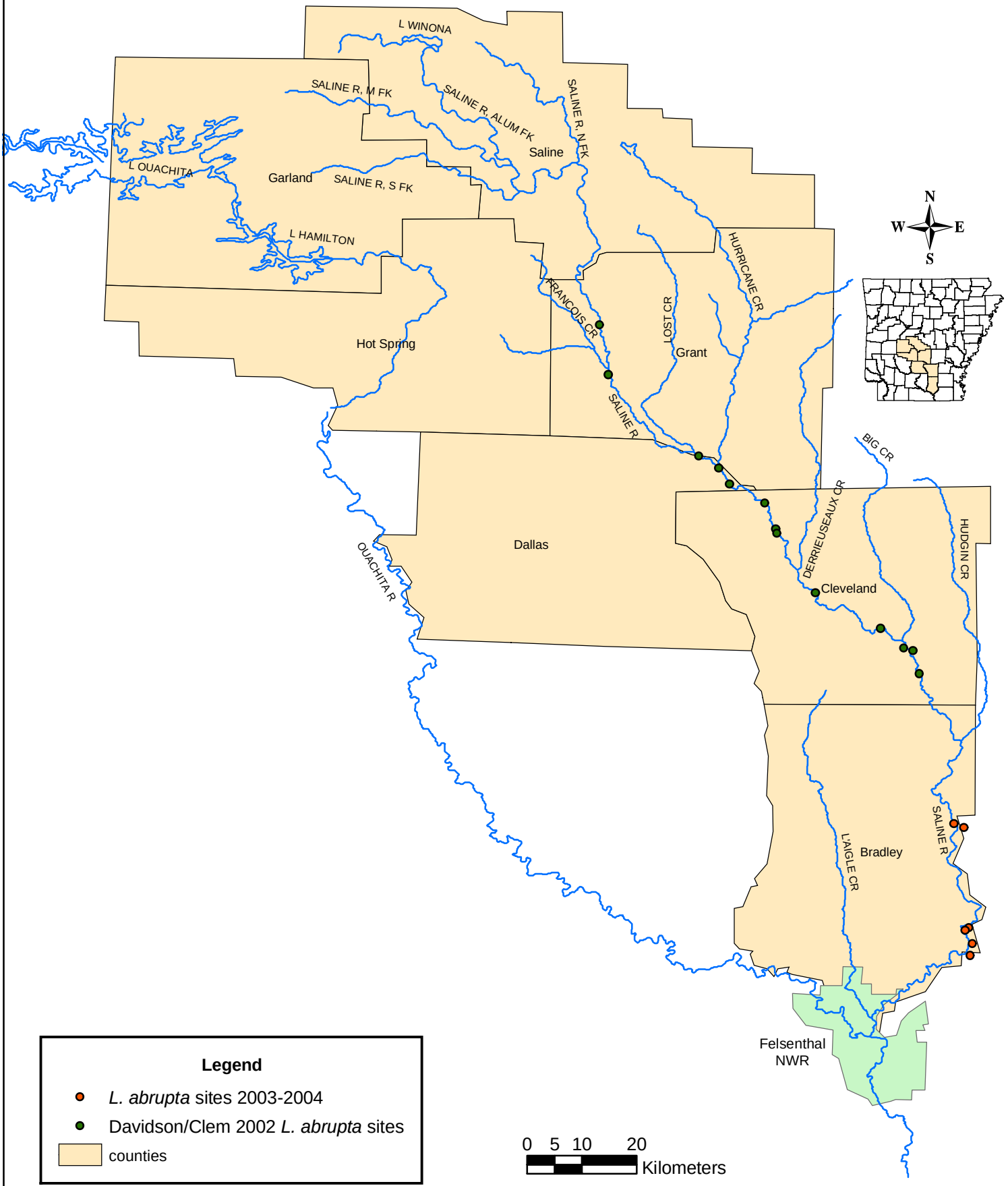
Lampsilis abrupta (Say 1831) – Pink Mucket. Distribution: Figure A3. Status: Federal – Endangered, TNC – G2, State – S2.

L. abrupta was collected from 6 sites between Arkansas Highway 15 and Felsenthal NWR (SRB128; SRC130; SRB141; SRB143; SRB147; SRB149). The maximum number of individuals sampled from a bed or concentration was three (SRB141), which was 1.4% of individuals examined from SRB141. *L. abrupta* population numbers appear unstable due to low numbers and limited dispersal in the Saline River. Davidson and Clem (2002) reported 11 sites for this species in the Saline River. This now brings the total identified sites for the pink mucket to 17 in the Saline River. Of these 17 sites only one apparent juvenile (4 to 7 years based on external annuli), was collected from SRB40. This may suggest that limited reproduction is occurring within the population.

Quadrula fragosa (Conrad 1835) – Winged Mapleleaf. Distribution: Figure A4. Status: Federal – Endangered, TNC – G2, State – S1.

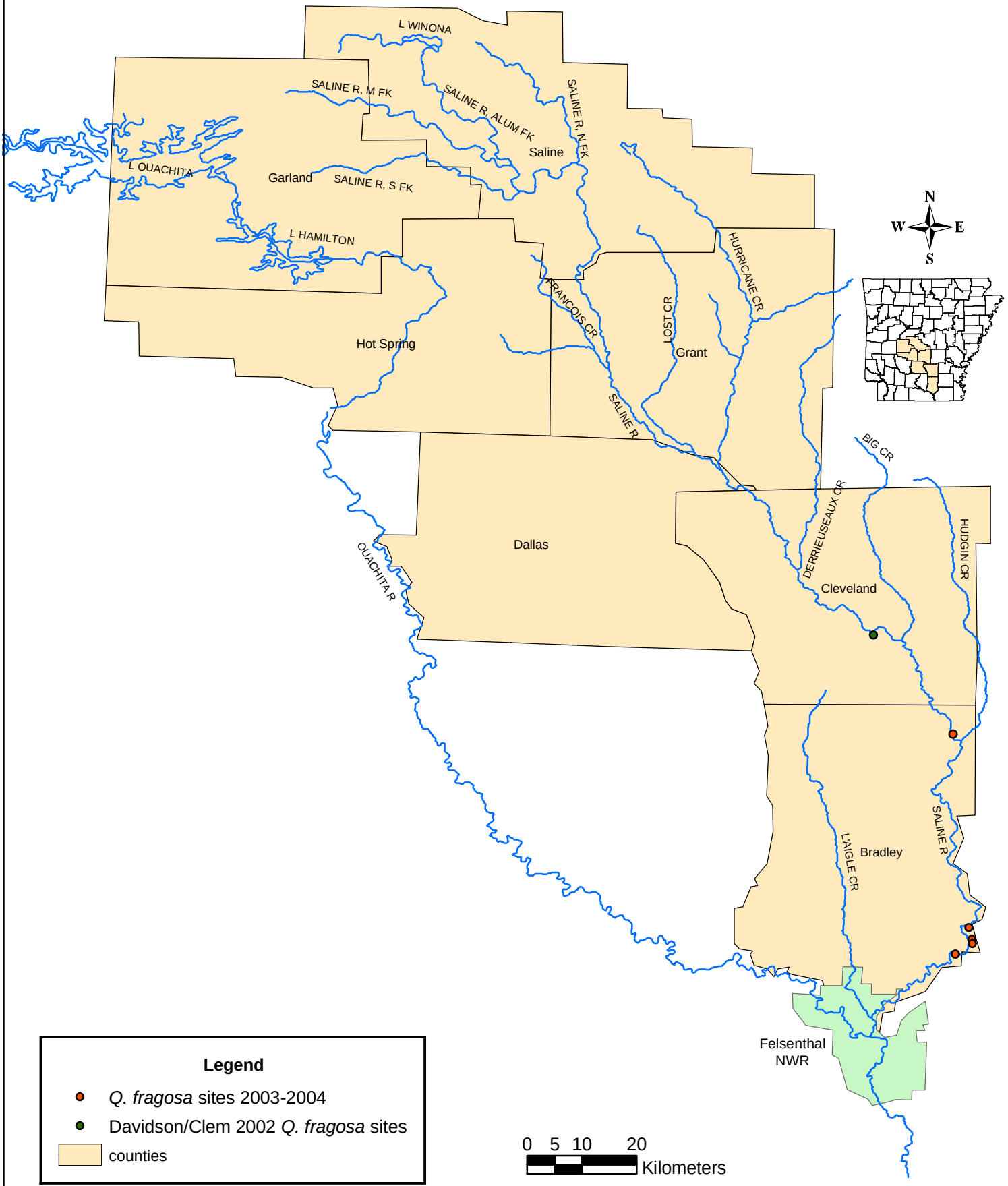
Q. fragosa was collected from six sites (SRB101; SRB141; SRB145; SRB146; SRB147; SRB152) within the survey area. The substrate consisted of gravel with sand. One juvenile (4 years old based on external annuli) was collected at SRB145. The collection of this individual would seem to indicate that there is some recruitment occurring in the Saline River. The total number of known locations in Arkansas for this species is 10, of which 7 exist in the Saline River. However, additional sites are suspected from the Ouachita River upstream of Camden, Arkansas.

Figure A3. *L. abrupta* sites in the Saline River from near Tull, Arkansas to Felsenthal NWR, 2001-2004.



Davidson and Clem, 2004

Figure A4. *Q. fragosa* sites in the Saline River from near Tull, Arkansas to Felsenthal NWR, 2001-2004.



Species of Global and State Concern

Cyprogenia aberti (Conrad 1850) – Western Fanshell. Distribution: Figure A5. Status: TNC – G2, State – S2.

> *C. aberti* is common (8th most common species) throughout the survey area, although it is less common downstream of Arkansas Highway 8 or Longview Landing. Two hundred seventy (270) individuals representing 2.7% of the total community were collected from 60 sites. *C. aberti* was present at 50 of 55 sites upstream of Longview Landing and only 10 of 28 sites downstream. The declining presence downstream probably is related to the occurrence of fewer riffles and subsequently a lower population of host fish (logperch). The species generally occurs in low numbers, but were locally abundant at several sites.

Ligumia recta (Lamarck 1819) – Black Sandshell. Distribution: Figure A6. Status: TNC – G5, State – S2.

> *L. recta* occurred throughout the survey area. However, it was restricted to eight sites (SRB101; SRC56; SRB103; SRB106; SRC57; SRB134; SRB164; SRB169). Relicts of this species also were collected from three additional sites (SRC58; SRB114; SRB116). This species appears to be unstable in the Saline River downstream of Arkansas Highway 15 due to low numbers and limited distribution.

Obovaria jacksoniana (Frierson 1912) – Southern Hickorynut. Distribution: Figure A7. Status: TNC – G1/G2, State – S2.

> *O. jacksoniana* occurs throughout the survey area (55 sites). *O. jacksoniana* is widely distributed in the Saline River, but is locally abundant only at sites downstream of Longview Landing. This species appears to be stable in the survey area.

Quadrula cylindrica (Say 1917) – Rabbitsfoot. Distribution: Figure A8. Status: TNC – G3, State – S2.

> *Q. cylindrica* was restricted to 9 sites (SRB99; SRB100; SRB117; SRB118; SRB128; SRB146; SRB147; SRB157; SRB163) widely distributed throughout the survey area. SRB146 had the greatest number of individuals (9) collected, which was 5.4% of the total community. This species is restricted to a two-mile reach upstream of Arkansas Highway 15 (Davidson and Clem 2002) and occurs in low numbers downstream of this point. The authors feel that *Q. cylindrica* population numbers are unstable throughout the Saline River.

Figure A5. *C. aberti* sites in the Saline River from near Tull, Arkansas to Felsenthal NWR, 2001-2004.

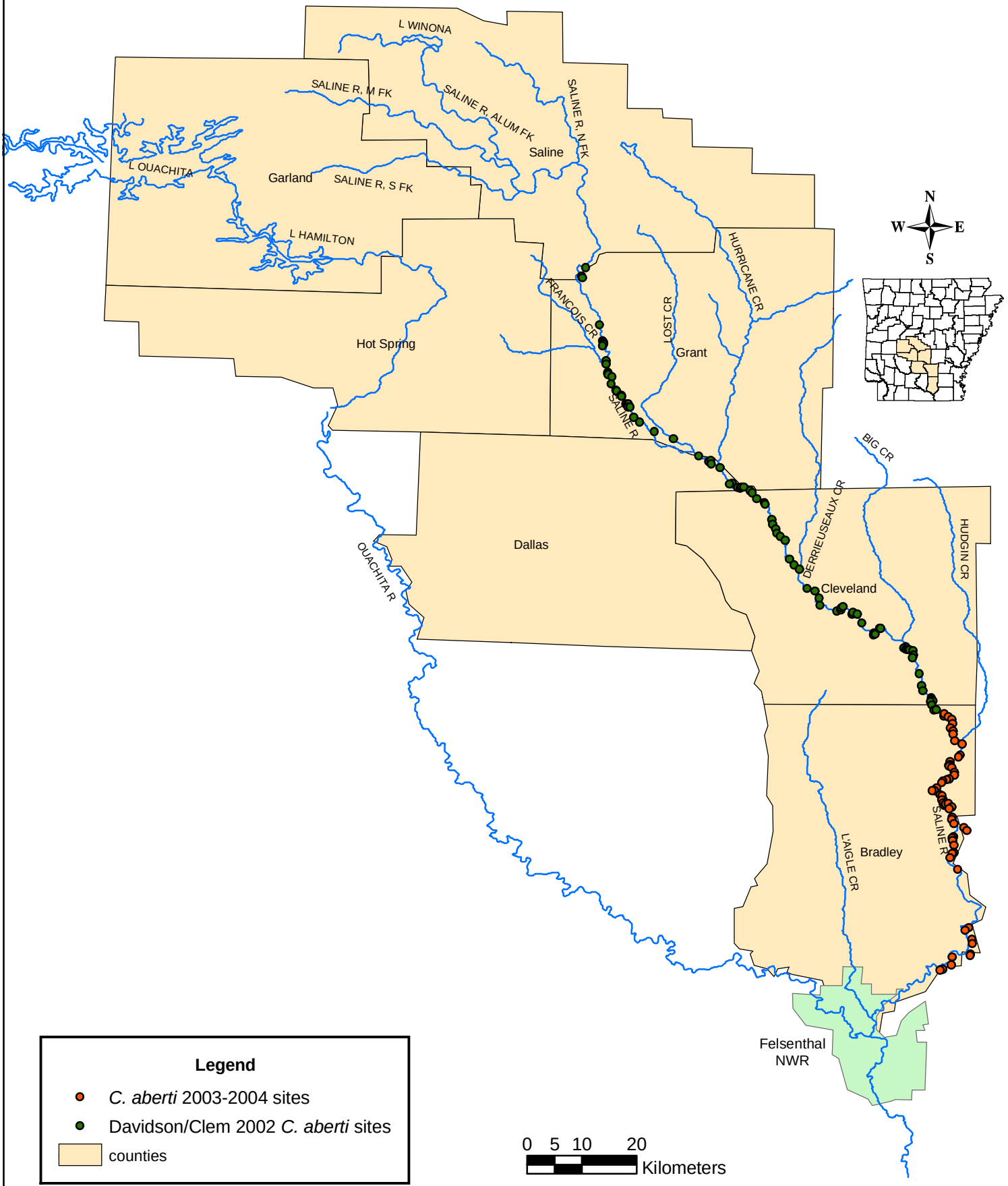


Figure A6. *L. recta* sites in the Saline River from near Tull, Arkansas to Felsenthal NWR, 2001-2004.

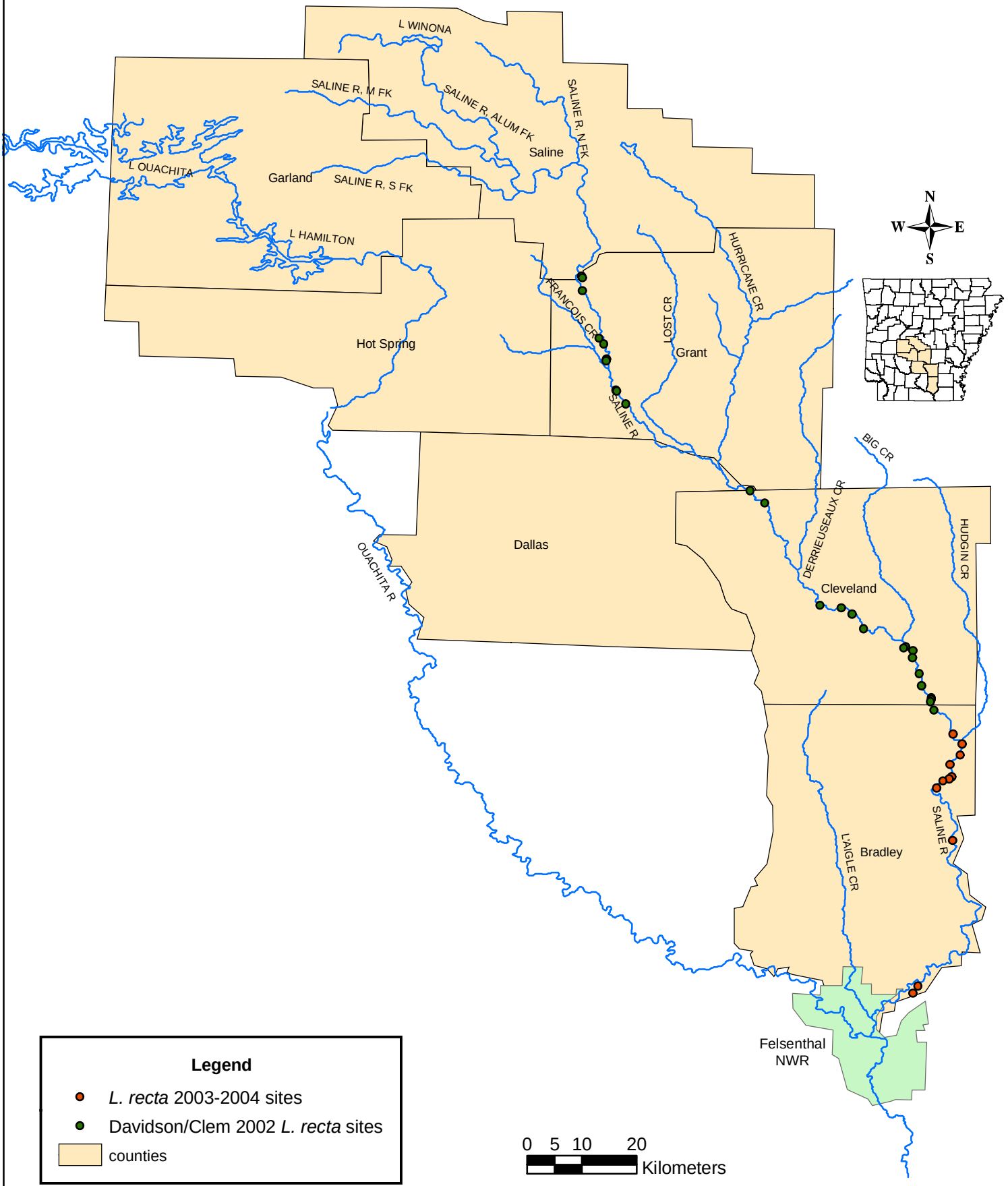


Figure A7. *O. jacksoniana* sites in the Saline River from near Tull, Arkansas to Felsenthal NWR, 2001-2004.

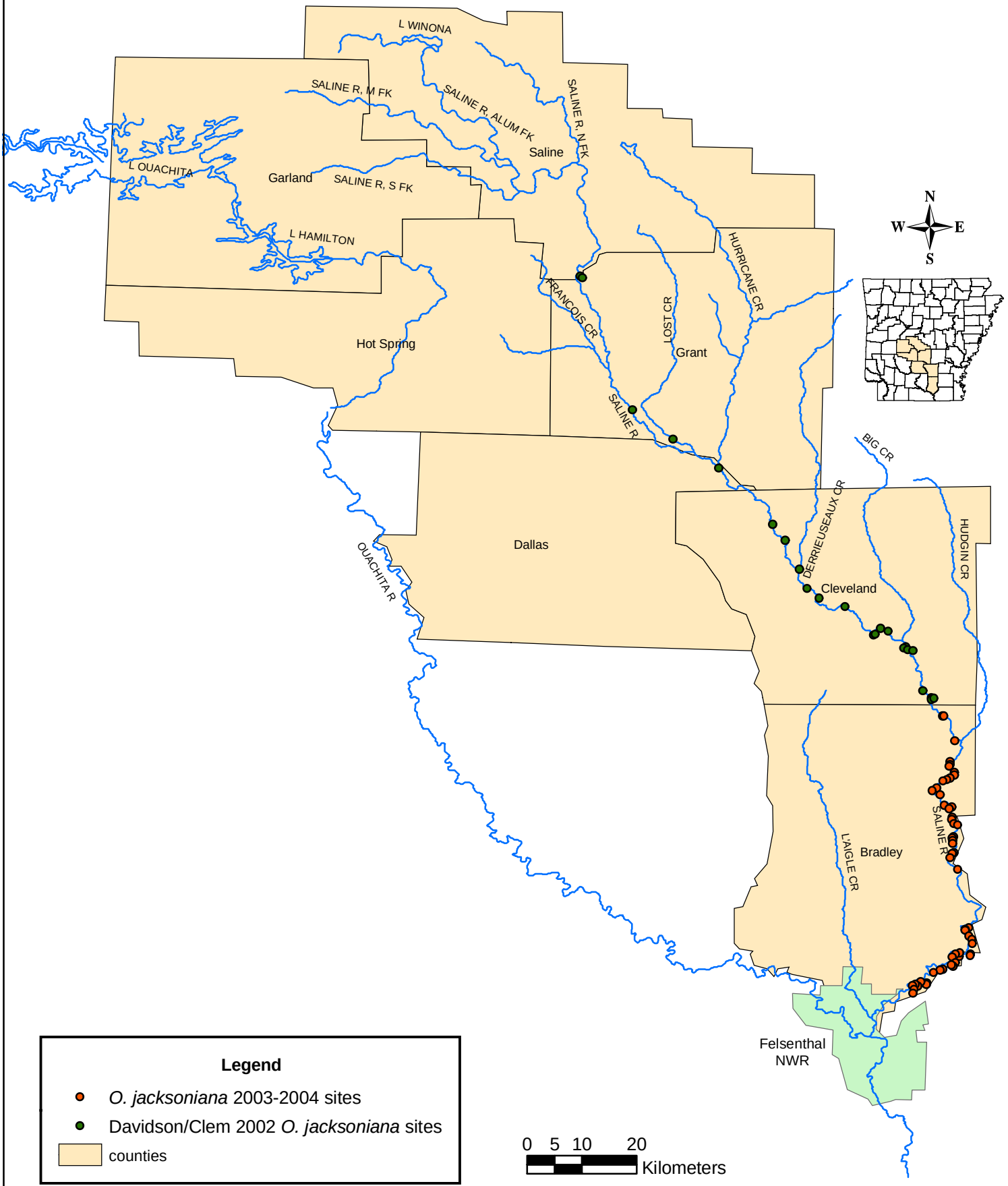
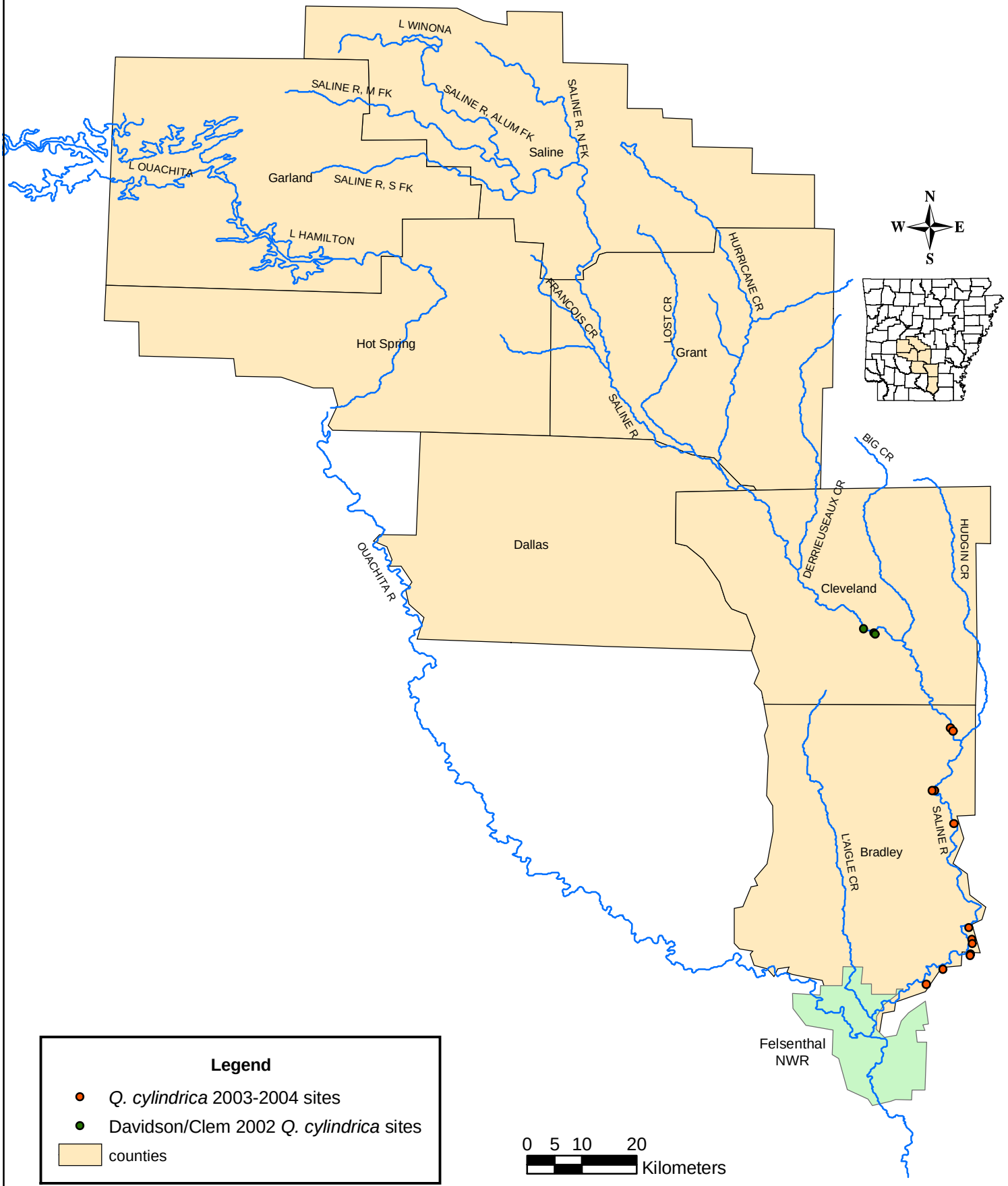


Figure A8. *Q. cylindrica* sites in the Saline River from near Tull, Arkansas to Felsenthal NWR, 2001-2004.



Davidson and Clem, 2004

Pleurobema rubrum (Rafinesque 1820) – Pyramid pigtoe. Distribution:
Status: TNC – G2, State – S2.

> *P. rubrum* was widespread throughout the study area. This species was the seventh most common (3.6%) species encountered in the survey area. *P. rubrum* dominated ten beds or concentrations upstream of Arkansas Highway 15, most (8) occurring between RMR Camp and Arkansas Highway 167 (Davidson and Clem 2002). This species is not as abundant or common downstream of Arkansas Highway 15. It also was not dominant in any beds or concentrations in this reach. This species appears to be stable within the Saline River, often occurring at high densities upstream of Arkansas Highway 15.

Quadrula apiculata – Southern Mapleleaf. Status: TNC – G2, State – S1.

Q. apiculata was collected from one site (SRB101). This species is extremely rare in the Saline River. Based on results from this survey and previous surveys, it is unlikely that a viable population exists.

DISCUSSION

Endangered, Threatened and Special Concern Species

Two federally listed endangered species, *Lampsilis abrupta* and *Quadrula fragosa*, were encountered during this survey from six new locations for each species. Seven species, *Cyprogenia aberti, Ligumia recta, Obovaria jacksoniana, Pleurobema rubrum, Quadrula cylindrica, Toxolasma lividus* and *Quadrula apiculata,* of global or state concern were encountered during this survey. Global and state concern species were common, and sometimes locally abundant, throughout the study area. A summary of each species distribution in the Saline River is provided in the results section.

Lampsilis abrupta is generally considered an inhabitant of medium to large rivers, and it is usually associated with mussel beds. This species prefers moderate to swift current over sand/gravel or gravel substrates. *Quadrula fragosa* is generally considered an inhabitant of medium to large rivers. The species prefers mud, sand and gravel substrates.

Management strategies should concentrate on maintaining the quantity and/or quality of habitat and augmenting populations throughout the study area in order to sustain viable populations. Long-term monitoring is needed to track population trends and determine successes/failures for population augmentation and reintroduction. Areas requiring special management attention are identified in the results section.

Additional Management Concerns and Needs

Protection or improvements in the status of mussels in the Saline River require proper management of the watershed and cooperative efforts of stakeholders. Private

landowners and state and federal agencies should consider efforts to stabilize existing habitats (i.e. stream bank stabilization projects) and sustain exceptional water quality. Execution of these management strategies should improve and/or sustain the quality and quantity of vegetative cover in riparian zones, decrease siltation and subsequently improve habitat quality for mussels.

ACKNOWLEDGEMENTS

The Arkansas Game and Fish Commission and U.S. Fish and Wildlife Service (Section 6 of the Endangered Species Act) provided funding for this survey. Joe Stoeckel, Arkansas Tech University, was integral in helping to secure funds.

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